



## Review

## Neurotoxicities induced by micro/nanoplastics: A review focusing on the risks of neurological diseases

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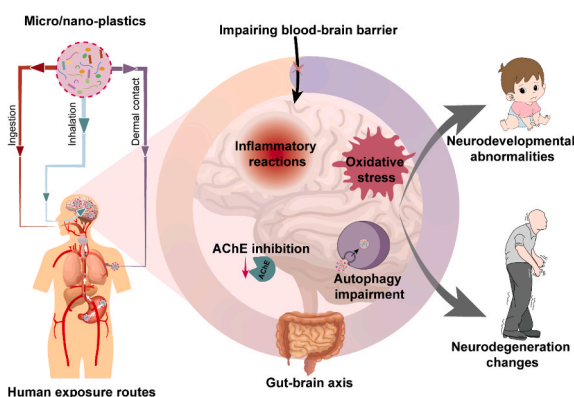
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## HIGHLIGHTS

- Environmental MPs/NPs can enter the human body and be transferred to the brain.
- MPs/NPs accumulated in the brain contributes to the development of neurological diseases.
- The neurotoxic mechanisms induced by MPs/NPs are summarized.
- Key challenges and further research directions of MP/NP-neurotoxicity are highlighted.

## GRAPHICAL ABSTRACT



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## ABSTRACT

Pollution of micro/nano-plastics (MPs/NPs) is ubiquitously prevalent in the environment, leading to an unavoidable exposure of the human body. Despite the protection of the blood-brain barrier, MPs/NPs can be transferred and accumulated in the brain, which subsequently exert negative effects on the brain. Nevertheless, the potential neurodevelopmental and/or neurodegenerative risks of MPs/NPs remain largely unexplored. In this review, we provide a systematic overview of recent studies related to the neurotoxicity of MPs/NPs. It covers the environmental hazards and human exposure pathways, translocation and distribution into the brain, the neurotoxic effects, and the possible mechanisms of environmental MPs/NPs. MPs/NPs are widely found in different environment matrices, including air, water, soil, and human food. Ambient MPs/NPs can enter the human body by ingestion, inhalation and dermal contact, then be transferred into the brain via the blood

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circulation and nerve pathways. When MPs/NPs are present in the brain, they can initiate a series of molecular or cellular reactions that may harm the blood-brain barrier, cause oxidative stress, trigger inflammatory responses, affect acetylcholinesterase activity, lead to mitochondrial dysfunction, and impair autophagy. This can result in abnormal protein folding, loss of neurons, disruptions in neurotransmitters, and unusual behaviours, ultimately contributing to the initiation and progression of neurodegenerative changes and neurodevelopmental abnormalities. Key challenges and further research directions are also proposed in this review as more studies are needed to focus on the potential neurotoxicity of MPs/NPs under realistic conditions.

## 1. Introduction

Plastic pollution has been becoming a global environmental problem to tackle. Current research indicates that the production and sales of plastics have grown exponentially since the latter half of the 20th century,[1] and approximately 79% ends up as contaminated garbage accumulated in landfills or the natural environment.[2] These discarded plastic debris are subjected to break into microplastics (MPs, defined as particles with a diameter less than 5 mm, including nanoplastics, NPs, plastic particles < 1  $\mu\text{m}$ ) through photodegradation, wave erosion, contact with animals, friction of sand and stone, water immersion and other processes.[3,4] Inevitably, MPs/NPs are ubiquitously found in various environment matrices.[5].

Ambient MPs/NPs enable to enter the human body and penetrate biological barriers, then spread throughout the body within the circulatory system.[6,7] The accumulation of MPs/NPs in tissues may cause a variety of toxic effects, leading to oxidative stress, immune inflammation reaction, cellular damage, and endothelial leakiness, as well as neurotoxic and metabolic disorders.[8] Despite the protection of the blood-brain barrier (BBB), the central nervous system (CNS) is vulnerable to MPs/NPs due to particles' special physical and chemical properties. Previous reports manifested that MPs/NPs can transfer into the brain and cause histopathological alterations, damage to the BBB, and impairment of neurological function,[9–11] which may ultimately increase the onset and development of neurodevelopmental and/or neurodegenerative diseases. Previous reviews have primarily emphasized on the general assessment of the risks to human body, with only a few dedicated to the neurotoxicities of MPs/NPs. These latter reviews just focused on summarizing the neurotoxic effects within different test subjects or intracellular behaviors of MPs/NPs.[12,13] There is still a need for a thorough evaluation on the hazards of neurological illnesses caused by MPs/NPs.

Literature published between 2000 and 2024 was obtained by searching for the terms “microplastics,” “nanoplastics,” “environmental risk,” “neurotoxicity,” “human exposure,” and “neurological diseases” in the online databases of Web of Science. We conducted a comprehensive review focusing on the neurotoxic effects of MPs/NPs on human health. The review highlights: (1) the environmental hazards and human exposure pathways associated with MPs/NPs, (2) the ways MPs/NPs move and spread in the brain, (3) the neurotoxic effects induced by MPs/NPs on neurodegeneration changes and neurodevelopmental abnormalities, and (4) the possible mechanisms of MP/NP-induced neurotoxicity. This will enhance comprehension regarding the extent to which MPs/NPs contribute to the onset of neurodevelopmental and/or neurodegenerative risks. Greater emphasis will be needed in plastic pollution and associated health problems not just for human health, but for sustainable development of the environment.

## 2. Environmental exposure of MPs/NPs to human body

### 2.1. Characteristics of environmental MPs/NPs

Environmental MPs/NPs can be classified into two categories, the primary and secondary. On purpose, the primary MPs/NPs are intentionally produced at the microscale for application in cosmetics, cleaning products, pharmaceuticals and air-blasting media.[14] These plastic

granules often comprise of polyethylene (PE), polypropylene (PP), polystyrene (PS) and polyolefin plastic spheres, with diameters varying from 1 to 5  $\mu\text{m}$ . Unlike the primary MPs/NPs, the secondary MPs/NPs are degraded from the large pieces of plastics undergoing ultraviolet radiation (UV), physical wear, and biodegradation.[15] The secondary MPs/NPs, which have a wider particle size distribution and manifest as fragments, fibers, granules, flakes and spheres, are the predominant source of MPs/NPs in the environment.[16].

### 2.2. Environmental MPs/NPs in air, water and soil

MPs/NPs are prevalent in various environmental compartments, including atmosphere, freshwater, wastewater, ocean, and soil (as shown in Table S1). The primary sources of atmospheric MPs/NPs are synthetic fabrics, degradation of synthetic rubber tires, and urban dust. Other significant contributors include building materials, industrial emissions, landfills, traffic particles and waste incineration.[17] These diverse forms of MPs/NPs are consistently transported and deposited in soil or sediment following their release into the air, potentially contributing to human respiratory exposure. As an example, the mean accumulation rates of MPs/NPs in urban and suburban regions of Paris, France constitute  $110 \pm 96$  and  $53 \pm 38$  particles per  $\text{m}^2/\text{day}$ , respectively.[18] The corresponding rates for Dongguan (China) and central London (United Kingdom) are  $36 \pm 7$  particles per  $\text{m}^2/\text{day}$ [19] and  $771 \pm 167$  particles per  $\text{m}^2/\text{day}$ , respectively.[17] Soil serves as an important reservoir for MPs/NPs,[20] which originate from various sources including the breakdown and deterioration of plastic films on agricultural land, fragmentation of plastic debris and plastic products in landfills, littering, atmospheric deposition, surface runoff, and irrigation with wastewater.[21,22] Multiple studies have determined a considerable quantity of MP fibers or debris in coastal, industrial, agricultural, greenhouse, and floodplain soils, with the variation in concentrations being closely related to human activities.[23] Studies have indicated that the building land in Beijing and the e-waste disposal zone soil in Guangdong contained a high abundance of MPs/NPs at 272 to 13,752 items/kg[24] and 0 to 34,100 items/kg, respectively.[25] However, in the Tibetan Plateau devoid of human activity, the abundance of MPs ranged from 20 to 110 items/kg.[26] MPs/NPs that are not completely encapsulated in sewage sediment or eliminated via filtration during the sewage treatment procedure, will inevitably enter freshwater systems. Rivers are estimated to contribute between 1.2 and 2.4 million tonnes of MPs/NPs, while water treatment facilities discharge between 50,000 and 15 million particles of MPs/NPs annually.[27] The ocean serves the most substantial receptacle of plastic waste, absorbing nearly 10% of the total global plastic production.[28] An excess of 8 million tonnes of plastics are estimated to infiltrate the oceans annually.[29] An instance of exceptionally high plastic waste concentrations can be found in the central North Pacific Ocean, where a region spanning approximately 1.6 million  $\text{km}^2$  in size contains at least 79 thousand tonnes of plastics, of which MPs/NPs make up 8%.[30].

### 2.3. MPs/NPs in food chains and human food

Plastics exhibit exceptional resistance to degradation and persist in the environment for an extended duration, which facilitates the entry of MPs/NPs into the human food chain. At the base of the food chain,

specifically in zooplankton such as copepod and ostracod, the occurrence of MPs/NPs has been documented.[31] Bivalves, such as blue mussels (*Mytilus edulis*) and Pacific oysters (*Crassostrea gigas*), own  $36 \pm 7$  and  $47 \pm 16$  particles/100 g, respectively.[32] MPs/NPs are frequently identified in fish tissues including fish intestines, stomach, and gills.[33,34] MPs/NPs also pervade in terrestrial food chains; they were found in chicken stomachs with a concentration of 5.1 particles/g [35] and sheep feces with 1000 particles/g plastic particles.[36] Contamination of MPs/NPs has been reported in other types of food, such as vegetables, fruits,[37] and staple food.[38] In addition, in our daily life, many processed foods, such as milk,[39,40] beer[39], table salt,[41,42] honey,[39] beverages,[43] and bottled water[44] have also been found to contain MPs/NPs. Further evidence suggests that the origin of MPs/NPs in food may be the packaging materials, verifying that processed foods are more likely to contain these substances than unprocessed foods.[45].

#### 2.4. Exposure routes of MPs/NPs to human body

As the ubiquitous environmental pollutants, MPs/NPs are capable of entering the human body via ingestion, inhalation, and dermal contact (Fig. 1).[46] Among these factors, ingestion is considered to be the major exposure route for MPs/NPs.[47] MPs/NPs have been widely detected in almost all sorts of food. On oral exposure alone, annual consumption of MPs/NPs per capita ranges from 39000 to 52000 particles.[48] Inhalation is a vital exposure pathway that cannot be ignored as well, contributing to nearly half of total human exposure. MPs/NPs in the ambient air are drawn primarily from synthetic textiles, rubber tires,

city dust plastic and abrasion of materials in buildings, waste, and landfills,[5,18,49] with detected concentrations varying from  $< 1$  to  $> 1000$  particles/ $\text{m}^3$  (outdoor) and from  $< 1$  to  $1583 \pm 1181$  (mean) particles/ $\text{m}^3$  (indoor), even consisting of inhalable sizes.[50] Especially, the rapidly-increased use of masks during the COVID-19 outbreak added more risks in respiratory tract exposure of plastic particles.[51] In a breathing-simulation experiment, researchers discovered that just a disposable medical mask enables a release of approximately  $1.1 \times 10^9$  particles with simulated breathing.[52] MPs/NPs interacting with the respiratory (and gastrointestinal) epithelium may diffuse or cellularly penetrate into the circulatory system and other organs.[5] Human intake of MPs/NPs has been estimated to be 74000 to 121000 particles per year when inhalation is considered.[48] Furthermore, despite a small percentage, dermal exposure may occur when human skin contact with personal care products or water containing MPs/NPs.[53] Skin is an important passive receptor for human exposure to MPs/NPs, and an Iran study with 2000 participants showed that more than 4000 MPs/NPs were adsorbed on both face and hand skin of a person during one day.[54] Taken together, it is estimated that on average, human can intake 0.1 to 5 g of MPs/NPs weekly through various exposure pathways, which imposes additional burdens on human body.[55] Inevitably, MPs/NPs have been found in various human specimens, including blood, feces, lung, and placenta, among others (as shown in Table S2).

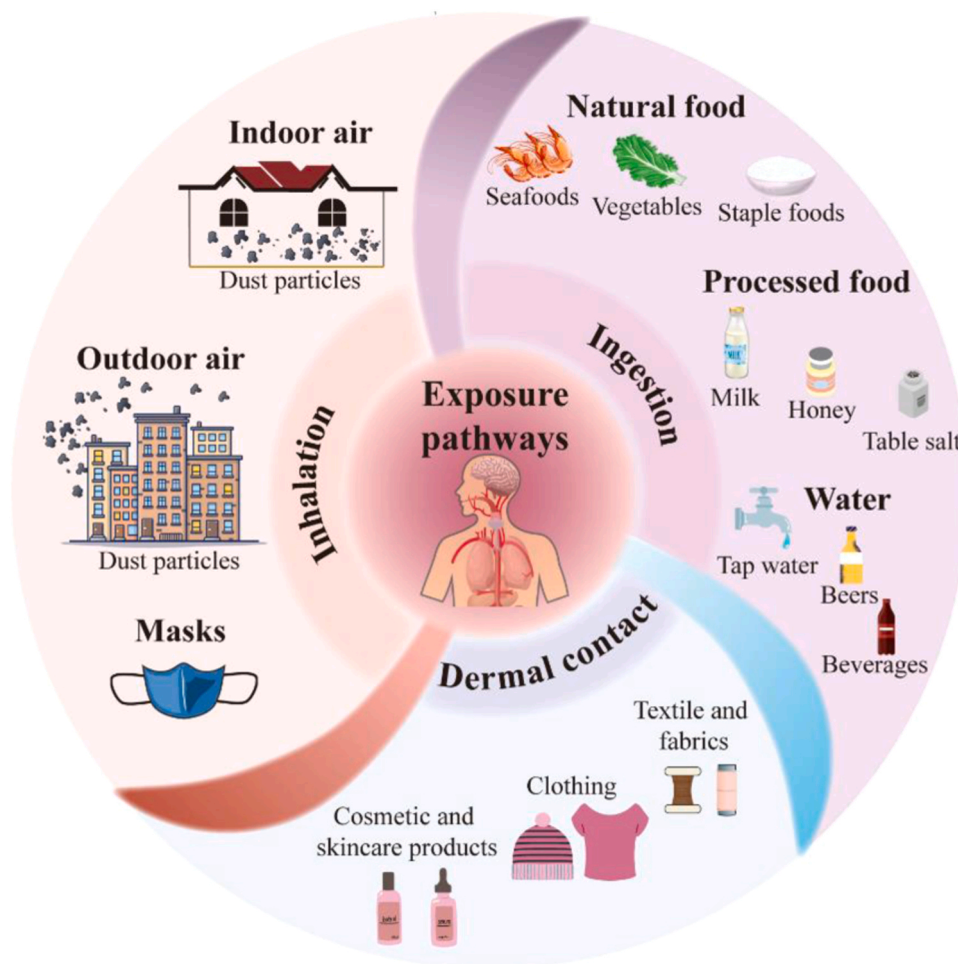


Fig. 1. Exposure pathways of environmental MPs/NPs to human.

### 3. Translocation and distribution of MPs/NPs in brain

#### 3.1. Translocation of MPs/NPs to the brain

The BBB is a protective barrier composed of endothelial cells that are tightly joined by junctions. Its function is to shield the brain tissue from potentially harmful substances present in the blood plasma.[56] Transporting across the BBB with blood circulation might be the first possible approach for the translocation of MPs/NPs to the brain (Fig. 2). Previous studies have pointed out that MPs/NPs, especially for those nanometer-sized particles, can penetrate the BBB, and then enter into the brain.[57–59] An *in vivo* test observed the accumulation signals of NPs in mouse brain tissues as early as 2 h after gavage exposure,[59] which revealed the rapidity of the transfer pathway. It is unclear the underlying mechanisms of MPs/NPs crossing BBB in the brain. Multiple studies have found that MPs/NPs may impair the tight junction structure that is consistent with the integrity of BBB, then increase BBB permeability, and lead to the transferring of particle pollution to the brain.[9, 58] But other besides, the molecular dynamics simulation provided a new train of thought for explaining the transport mechanism. According to recent research in Austria using computer models, the presence of a biomolecular corona around plastic particles is crucial for their ability to pass across the BBB. The study found that cholesterol molecules can boost the uptake of these pollutants into the BBB membrane.[59]

Particles inhaled through the nasal route might flee from blood circulation and directly be transported via nerve pathways (Fig. 2).[60,61] The existing studies on inhaled MPs/NPs are scant and only one study reported that the inhaled NPs could tend to aggregate in the brain rather than the liver, compared with NPs injected by a tail vein, indicating the expedited delivery between brain and nose. Researchers hypothesized, on the basis of *ex vivo* fluorescence image results, that NPs could be transported to the olfactory bulb and cerebral cortex via the olfactory nerve pathway, as well as the cerebellum and pons via the trigeminal pathway.[10] Despite little data on the olfactory nerve pathway and

trigeminal pathway of MPs/NPs transport, typical nanoparticles, such as metallic oxide,[62,63] ultrafine elemental  $^{13}\text{C}$  particles,[64] carbon black particles,[65] have been proven to deposit in mammalian brains by nasal inhalation. Therefore, more and more studies will be needed to verify the transferring mechanism involved the inhalable plastics to the brain.

#### 3.2. Distribution of MPs/NPs in brain tissue

MPs/NPs entering into the brain with different approaches might tend to be deposited in different sites. Studies based on rodent models have shown that uptake of MPs/NPs by oral gavage or drinking water could be transported along the blood flow to all parts, especially in the middle of brains.[58,66,67] For example, a noticeable emission of light was observed in the central region of mouse brains that were orally administered with PS-NPs (50 nm) labeled with crimson red, and the intensity of fluorescence increased in proportion to the dosage.[58] Similar results were also found in mice exposed to PS-MPs (5  $\mu\text{m}$ ) by drinking water. Animal imaging analysis showed that MPs were distributed throughout the brain, and even the cortex and hippocampus, after 3-day administration of MPs.[66] MPs/NPs transported through nerve-to-brain pathways may target the olfactory bulb, cortex, cerebellum, and brainstem, with distinct nerve pathways leading to various principal deposition areas.[68] As an illustration, the olfactory bulb is the location where particles are deposited through the olfactory nerve-to-brain transport pathway[69], while the taste nerve-to-brain pathway delivers MPs/NPs to the cortex, which primarily accumulate in that area.[70] Inevitably, MPs/NPs are able to deliver from the initial sites to other areas of the brain over time. Supporting evidence for this was discovered in the study, which demonstrated that the inhaled NPs were transported from the olfactory bulb to the cerebrum, cerebellum, and basal forebrain.[10].

The factors influencing the deposition of the MPs/NPs delivered into brain are diverse. The type of MPs/NPs was reported to be as a crucial

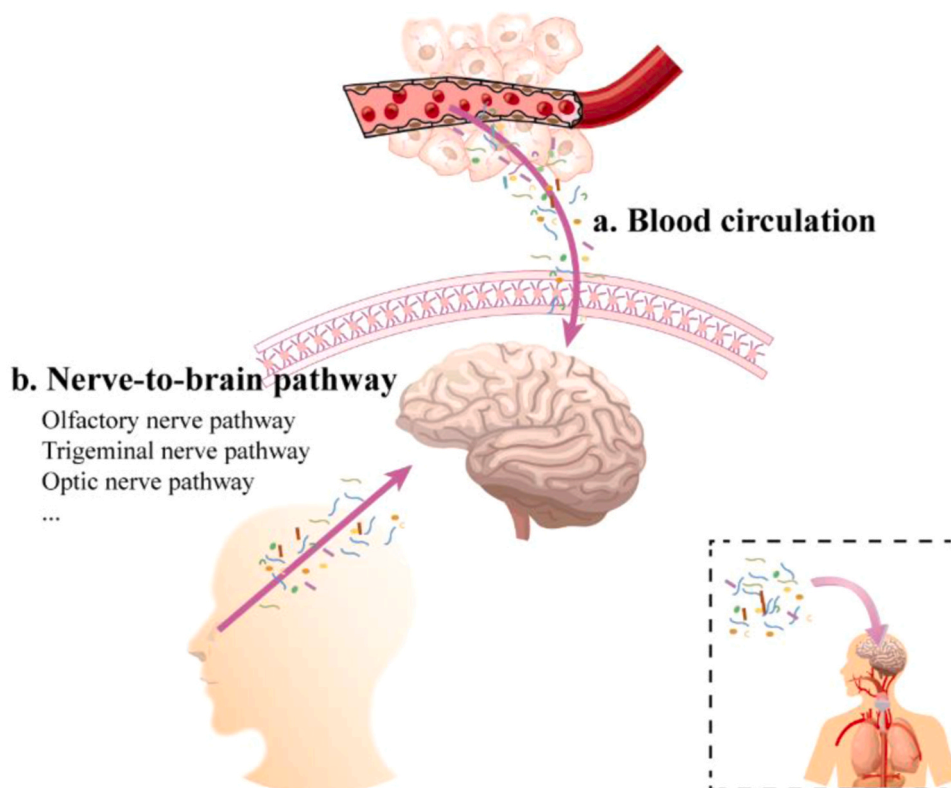


Fig. 2. Translocation pathways of MPs/NPs to the brain.



determinant. Few toxicological evaluations were tested except for PS-MPs/NPs, and currently most studies were performed to be modeled by commercialized PS microspheres. However, different types of MPs/NPs might possess varying depositional characteristics due to particular material properties. As an illustration, with similar administration manner and dosage, it was found that PS microspheres were more easily gathered in the deeper of the brain than PP. [71] Due to no more research, we speculated that the cause of the discrepancy was that PS owns the higher density. Furthermore, the size is another significant element contributing to the accumulation of MPs/NPs in brain. Smaller MPs/NPs are more readily accumulated in tissues than larger ones. [67, 70, 72] After the inhalation of 80 and 200 nm NPs, the deposition of NPs in mouse brains exhibited a size-dependent pattern: 80 nm PS-NPs were found in the olfactory bulb, cortex, and hippocampus, as well as deeper regions of brain such as the cerebellum and brainstem, whereas 200 nm PS-NPs were only found in olfactory bulb and cortex. [10] In addition, the dosage also exerts a significant influence in this process. One study provided evidence for the dose-dependent pattern for accumulation of MPs. [9] Following a continuous intake of drinking water containing PS-MPs for 180 days, the accumulation of microplastics in brain with high concentration (1000 µg/L) displayed a broader scale than that with low dose (100 µg/L). [9].

#### 4. Neurotoxic effects induced by MPs/NPs

##### 4.1. Neurodegeneration changes

Neurotoxicity induced by environmental factors is recognized as a key contributor to the pathogenesis of neurodegenerative diseases. Based on epidemiological and toxicological data, the section will introduce the neurodegenerative changes caused by MPs/NPs, including abnormal accumulation of protein, neuronal loss, and cognitive deficits, among others.

##### 4.1.1. Aberrant protein folding and amyloidosis

MPs/NPs may affect protein folding and stimulate the production of abnormal amyloid proteins, thus potentially initiating the onset of amyloidosis, a prominent hallmark of neurodegenerative disorders. Misfolded proteins form amyloid fibrils or large oligomeric aggregates during the dynamic process of protein fibrillation. Although the precise mechanisms underlying neurodegenerative disorders like Alzheimer's disease (AD) and Parkinson's disease (PD) remain unknown, it is hypothesized that the aggregation of misfolded proteins is the cause of nearly all of these diseases. While it is commonly accepted that these disorders typically arise randomly or as a result of hereditary causes, they have also been associated with other environmental and behavioral factors. Epidemiological studies across countries support the strong associations between the risk of neurodegeneration changes and particle exposures. One study reported that the abnormal aggregation of microtubule-associated protein tau and  $\alpha$  synuclein in the brainstem of 186 elderly residents of Mexico City was associated with the profuse presence of exogenous particles. [73] The process of fibril production initiates with a gradual contact between misfolded proteins and preexisting oligomers, which act as nuclei for the elongation of fibrils. The kinetics of protein fibrillation can be divided into three phases: the initial lag phase, the subsequent elongation phase, and the final saturation phase. During the lag phase, the creation of a nucleus is the crucial step that determines the rate of fibrillation. [74] Linse et al. showed *in vitro* that NPs increased the likelihood of seed formation for fibril nucleation, utilizing human  $\beta$ 2-microglobulin. They noted a reduced lag phase when NPs were present, and a correlation with the quantity of particle surface. An additional report investigated the insulin fibrillation induced by PS-NPs. [75] The findings indicated that the inclusion of NPs reduced the duration of the latency phase associated with insulin fibrillation; however, no substantial impact was observed on the remaining two phases. [76] Back to human health, as evidence suggests

that NPs are capable to disrupt the folding of native proteins and interact with intracellular amino acids, it is possible that they could initiate the clustering of amyloid proteins, which in turn lead to the formation of the early stages of AD. The hypothesis that environmental factors can act as triggers for AD is substantiated by the very low prevalence of genetic instances of AD and may elucidate, to some extent, the sporadic manifestations of this neurodegenerative disorder. [77] The alterations in abnormal protein folding and amyloidosis have the potential to heighten the susceptibility to neurodegenerative disorders.

##### 4.1.2. Neuronal loss

Neuronal loss is limited in mature CNS, although there may be noticeable variations in the number of neurons between young and old persons in specific brain regions. [78] Nevertheless, a considerable surge in neuronal depletion is observed in numerous neurodegenerative disorders when compared to age-matched controls; this finding is consistent with longitudinal investigations of the progression of the disease. [79, 80] A growing body of research found that the occurrence of neuronal loss may be prompted by environmental MPs/NPs. Some results based on pathological examination confirmed the above hypothesis. After being administered PS-MPs intranasally in Kunming (KM) mice for 4 weeks, the nerve cells in the hippocampus exhibited signs of disorder and laxity, a reduction in the amount of Nissl bodies, dispersion of pyramidal neurons, and a diminished or absent main dendrite structure within the brain. [81] Similarly, examination of brain tissue with hematoxylin-eosin (H&E) staining revealed a more intense staining of the nucleoli in neurons and a lower number of neuronal cells in the hippocampus of C57BL/6 mice subjected to MP/NP-PS plastic spheres. [9].

Another study found that exposure to MPs alone resulted in a significant increase in intercellular space, while mice after joint exposure to di-(2-ethylhexyl) phthalate (DEHP) and MPs had deeper nuclear staining, concentrated cell necrosis, and tangled neuronal fibers. [82] After being exposed to PS-MPs, the number of Purkinje cells in the cerebellar tissue of chickens was reduced, the gap between them and the granular layer was widened as observed by H&E staining. Additionally, cellular mitochondria were smaller, membrane density was increased, and cristae were reduced, as observed using transmission electron microscopy. [83].

Beyond that, certain neuronal loss has also been reported in MP/NP-exposed brains, especially cholinergic and dopamine (DA) neurons. Cholinergic neurons are widely distributed in brain regions and depend on acetylcholine (ACh) for their neurotransmission conduction. [84] The loss of cholinergic neurons in the nucleus basalis of Meynert in AD patients' brains is an essential cause of their progressive cognitive decline. [84, 85] After exposure to 0.5 and 1.0 µm PS-MPs/NPs, obvious damage in the cholinergic neurons was found, which was visualized by the significant decreases in fluorescence intensity of green fluorescence protein expression in cholinergic neurons of the transgenic nematode (*Caenorhabditis elegans*). [86] Another study confirmed that the cholinergic system alteration was mainly responsible for the neurotoxicity after exposure of zebrafish (*Danio rerio*) to waterborne PE-MPs. [87] The development of PD is believed to be triggered by the degeneration of around 80% of dopaminergic neurons in the substantia nigra region of the brain, resulting in a subsequent reduction in DA levels. Evidence suggested that exposure to MPs/NPs contributed to the occurrence of dopaminergic loss. PS-NPs may induce PD-like neurodegeneration and DA neuron deficit in the brains of mice primarily by causing mitochondrial dysfunction and disruption of energy metabolism. [88] The concentration of DA in the brain decreased gradually with decreasing particle size throughout the exposure process. Modified behavioral activity in the goldfish (*Carassius auratus*) was ascribed to the suppression of DA. [89].

##### 4.1.3. Abnormalities in neurodegenerative and affective behaviors

Neurobehavioral disorders encompass a wide range of behavioral

impairments that occur in conjunction with brain diseases, persistent brain impairments, and traumas. These disorders include affective disorders, personality disorders, and cognitive problems, among others. Anxiety and depression-like behaviors are some of the most prevalent and complex affective disorders induced by MPs/NPs. To illustrate, male C57BL/6 J mice assigned to the PS-MP/NP group exhibited elevated anxiety behavior, as evidenced by a substantial reduction in both the percentage of distance and time spent in the central area during the open field test. The percentages of time spent in the open arms were significantly reduced in the PS-MP/NP groups compared to the control group in the elevated plus maze test.[90] After chronic exposure to naturally weathered MPs, adult zebrafish displayed increased depression-like behaviors with faster delays in reaching the dark zone and longer stays in the dark zone in the white/dark test.[91] Moreover, antidepressant amitriptyline can ameliorate MP-induced anxiety-like behavior and adverse effects in juvenile brown trout (*Salmo trutta f. fario*), demonstrating the intimate relations between MPs/NPs and affective disorders.[92].

Cognitive deficit was also frequently found in MP/NP-treated animals. The water maze test revealed that 3 weeks of PS-MP exposure impaired the learning and memory functions, exploratory, and learning capabilities in KM mice.[81] The administration of PS-MPs to mice for 8 weeks was also linked to significant impairments in learning and memory behavior, as well as hippocampal injury.[93] The abnormal behavior induced by MPs/NPs is more evident with longer treatment duration. Following a continuous 180-day exposure to drinking water containing PS-MPs, mice displayed concentration-dependent cognitive and memory impairments as compared to control mice, as determined using the Morris water maze and novel object recognition tests.[10] Interestingly, several studies modeled by honey bees reported that acute exposure to poly (methyl methacrylate) (PMMA)-, PS-, and PE-MPs impaired bee learning formation and memory retrieval,[94,95] adding to evidence that plastic pollutants affect brain health and memory.

Furthermore, multiple disorders in neurobehaviors occur simultaneously in some situations. Acute exposure to PE-MPs resulted in abnormal behaviors such as chasing, stranding, seizures, and labored breathing in adult zebrafish, and the severity and percentage of fish showing behavioral changes increased with the concentration of MPs.[96] Chronic exposure to PS-NPs reduced locomotor activity, aggression, and predator avoidance meanwhile, dysregulated circadian rhythmic locomotor activity in adult zebrafish.[97] Neurobehavioral disorders are the external manifestation of the internal damage of the nervous system. These changes in behavioral function hint at brain health problems caused by MPs/NPs.

## 4.2. Impaired neurodevelopment

Neurodevelopmental diseases encompass a range of conditions that hinder the growth and development of the brain. These problems can arise from several factors, such as genetics, lesions, and environmental influences. Examples of neurodevelopmental disorders include intellectual disability (ID), autism spectrum disorder (ASD), and attention-deficit/hyperactivity disorder (ADHD), among others.[98] Neurodevelopmental diseases arise from disruptions in the growth of the nervous system, resulting in dysfunction at the neuronal, sensory, or behavioral levels. As the developing brain lacks a fully functional BBB, it is more susceptible to environmental MPs/NPs. Such exposure can induce tissue inflammation and exacerbate damage to vulnerable regions of the brain. In this section, the risks of neurodevelopmental negative outcomes caused by MPs/NPs will be discussed in detail.

### 4.2.1. Inhibited neuronal development

The development process of neurons is extremely complex, encompassing multiple stages such as cell differentiation, growth, axon extension, and synapse formation. The embryos' optical transparency and the ability to create transgenic zebrafish enable the observation of

individual neurons in zebrafish throughout embryonic development,[99] providing more evidence for MP/NP-induced abnormalities in neuron development. A study utilized transgenic zebrafish embryos to examine the impact of PS-NPs with varying particle sizes on motor neurons. The findings demonstrated that the motor neuron axon growth in transgenic zebrafish was substantially impeded by PS-NPs with all particle sizes, including axon loss, shortened or irregular axon length, and reduced synaptic density. These results suggest that the presence of PS-NPs directly induces motor neuron axonal developmental damage.[100] Zhang et al. demonstrated that PS-NP treatment markedly reduced the axon lengths of primary motor neurons and decreased expression of neuron outgrowth-related gene *ISL LIM homeobox 2a* (*isl2a*) in zebrafish.[101] In addition, some studies showed MPs/NPs made a profound impact on the expression of genes regulating synaptic/axonogenesis, including sonic hedgehog signaling molecule *a* (*shha*), growth-associated protein 43 (*gap43*), and  $\alpha$ 1-tubulin, suggesting axonal damage and disruption of neuronal formation, regeneration, and plasticity induced by MPs/NPs.[100,102–104] The study of transcriptomic analysis revealed that exposure to PE-MPs had an impact on the transcripts related to the normal development of brain structures, such as dendrites and nerves, as well as the functioning of synapses, including processes like vesicle cycling, vesicle-mediated transport in synapses, and synapse construction.[105].

### 4.2.2. Neurotransmitter disturbances

Neurotransmitters are naturally occurring substances that serve as messengers in the transmission of signals within cells across synapses in the CNS, mainly including monoamine neurotransmitters, amino acid neurotransmitters, and choline, among others.[106] It is generally admitted that the balance between the excitatory and inhibitory processes of neurotransmitters determines neuronal activity[107], and some neurological and psychiatric disorders are associated with abnormal production or function of neurotransmitters.[108].

Recent reports suggest that exposure to MPs/NPs can induce the alteration in neurotransmitter systems. A previous study found that PS-MPs dysregulated the cholinergic, dopaminergic, and GABAergic ( $\gamma$ -aminobutyric acid) neurotransmission systems, which reduced the swimming competence of zebrafish larvae.[109] Homologous changes were also found in another study, accompanied by the variational serotonin (5-HT), neuraminic acid (NA), DA, and ACh in zebrafish brains subjected water- and food-borne exposure to PE-MPs.[87] Another study examining the negative impacts of charged PS-MPs reported that exposure to these particles can disrupt the levels of certain neurotransmitters (such as glycine, cysteine, glutathione, and glutamic acid) in zebrafish larvae, leading to impaired neurobehavioral function.[110] Noteworthy, the neurotoxicity of aged MPs/NPs is more of a concern. Several studies have proven that UV-aged PS-MPs induced more severe impacts on neurotransmitter content and related gene expression than virgin MPs [104,111] owing to the extra adsorption of metals and other organic compounds.[112] Changes in the contents of neurotransmitters (ACh, DA, 5-HT, and GABA) and neurotransmission have also been found in other organisms, including goldfish, seabream (*Sparus aurata*), and blood clams (*Tegillarca granosa*).[89,113,114] Furthermore, the damage to neurotransmitter systems can be transferred generationally by maternal breast milk to progeny. The breastfed progeny whose mothers exposed to PS-NPs displayed an imbalance of glutamate/GABA in the cortex and the hippocampus.[115].

### 4.2.3. Abnormalities in neurodevelopmental behaviors

Early-stage CNS development is susceptible to environmental perturbations. Since locomotor activity is 10–100 times more sensitive than mortality, it can frequently be used as an endpoint for neurotoxicity.[116,117] PS-NPs with sizes of 100, 500, and 1000 nm were shown to hinder the movement of zebrafish larvae, indicating that plastic particles at the nanoscale can all lead to decreased neurological function.[100] Similarly, researchers demonstrated that PS-NPs can be accumulated in

the brains and sensory organs of zebrafish embryos. Low concentrations of NPs caused larvae to elicit hyperactivity in combination with increased anxiety, while activity was significantly reduced at high concentrations.[118] Abnormal behaviors like bursting and freezing appeared to increase, accompanied by seizure-like electroencephalographic signals in PS-MP treated zebrafish embryos.[109].

Recent studies have indicated MPs/NPs could represent a possible risk factor for the induction of ASD-like behaviors. Pre/post-natal exposure to PE-MPs exhibited more repetitive and compulsive behaviors in nestlet shredding and marble burying. It also led to a decrease in social interaction and interest in the three-chamber test, revealing ASD-like traits in mice.[119] Another study revealed that consuming PE-MPs led to a decrease in defensive social grouping and a decline in risk assessment behavior.[120] So et al. found that mice that were subjected to PS-MPs from embryonic to adulthood exhibited diminished interest in a familiar stranger compared to a novel stranger in the three-chamber social test. However, no alterations were observed in the emotional, compulsive, or cognitive behaviors.[121] These results suggest that MPs/NPs induced diminished social memory and cognition, as well as a lack of interest in social novelty.

In addition, other abnormalities in neurodevelopmental behaviors have been reported, such as cognitive deficit, emotional disorder, and sleep/wakefulness. As an essential characteristic that enables individuals to distinguish and favor particular objects, color preference has been documented to influence the memory and learning processes of zebrafish.[122] Zebrafish larvae exposed to 50 nm PS-NPs exhibited color preference confusion in comparison to those in the control group, indicating that PS-NPs had an impact on the CNS of the zebrafish.[123] One study based on the generational transfer of particles addressed that maternally ingested PS-NPs can reach the brain of progeny via breast milk and cause cognitive deficit by dysregulating brain sexual dimorphism in female mice.[115] Exposure to polyglycolic acid (PGA) and polylactic acid (PLA) as degradable MPs reduced voluntary movement, induced anxiety-depression-like behaviors, and disrupted sleep/wakefulness behaviors in zebrafish larvae.[124] Table 1.

## 5. Possible mechanisms of neurotoxicities induced by MPs/NPs

### 5.1. Undermining the integrity of BBB

As the primary protective barrier for neurons in the brain, the BBB is the basis for the brain's defense against systemic toxins and the maintenance of basic homeostasis required for neuronal functioning.[134] The BBB, which prevents external contaminations in the bloodstream from crossing to the brain, is an important line of defense against brain damage. However, some studies indicate that the BBB is vulnerable when faced with withering attacks from a wealth of particulate matters. For example, one study evaluated the permeability of BBB using Evans blue dye in MP-exposed C57BL/6 mice and found an increase of BBB leakage in PS-MP treated mice.[93] Similar results were observed in another study on PS-NPs in which fluorescein sodium and platelet endothelial cell adhesion molecule-1 (PECAM-1) were applied to assess BBB permeability.[58] PECAM-1 is a protein that is predominantly found at the junctions of endothelial cells in blood vessels and is frequently used as an indicator of the integrity of the BBB.[135] One hour prior to sacrifice, mice were injected intraperitoneally with 2% fluorescein sodium. Extravasated fluorescein, as well as decreased expression of PECAM-1 shown as immunofluorescence staining in brain cryosections were observed from the PS-NP treated mice, suggesting PS-NPs induced damage to mouse BBB.[58] Furthermore, the study provided more compelling evidence of the effects of MPs on the BBB integrity.[9] In their study, a biotin tracer assay was employed to assess the structural integrity of the BBB qualitatively. Changes in the ultrastructure of the BBB were also evaluated; and TEM images displayed the tight junction structure became shorter and thinner in mouse brains after chronic exposure to PS-MPs.[9] In addition, exposure to PS-NPs

significantly reduced the expression of tight junction proteins in the BBB of chickens, suggesting that the BBB in the cerebellum is disrupted.[11] After 7 days of exposure to 10 mg/L PS-NPs, particles were observed to accumulate in the blood and brain, indicating that the NPs penetrated the BBB and eventually reached the brain.[136] The increased permeability of BBB may impair the exchange of substances between the brain and blood, then perturb the internal environment homeostasis of CNS.[137,138].

### 5.2. AChE inhibition

Acetylcholinesterase (AChE) is a key enzyme responsible for breaking down ACh into choline and acetic acid. It plays a crucial role in maintaining the proper functioning of the nervous system. Since ACh primarily acts as a neurotransmitter that regulates motor neuron function, inhibiting the action of AChE is a notable indicator of neurotoxicity. This inhibition leads to a substantial increase in ACh levels in the brain, ultimately disrupting the functioning of the nervous system.[139, 140] Moreover, the buildup of ACh in the synaptic cleft causes excessive activation of receptors, hinders the transfer of neurotransmission, and can ultimately end in paralysis and mortality. Many studies have recently demonstrated the inhibition of AChE caused by exposure to MPs/NPs, and it seems to affect a wide range of taxa, including invertebrates and mammals.[72,141] To illustrate, in the exposed group of PS-NPs, significant inhibition of AChE activity and gradual increase in the ACh level were detected in Javanese medaka fish (*Oryzias javanicus*). [142] According to a study by Ding et al., the accumulation of 0.1  $\mu$ m PS-MPs in the brain of red tilapia (*Oreochromis niloticus*) resulted in a substantial decrease of 37.7% in AChE activity, implying that PS-MPs have the potential to inhibit the catalytic capacity of AChE and could potentially induce neurotoxicity in fish.[143] The inhibitory effects of MPs/NPs on AChE had been also identified in another marine organism of discus fish (*Symphysodon aequifasciatus*), with the activity in the head being inhibited by MPs, which may be adverse to nervous and neuromuscular function, resulting in reduced predatory performance.[144] In addition, the neurotoxic properties of MPs/NPs have also been addressed in zebrafish. Torres-Ruiz et al. reported that AChE activity is inhibited by exposure to PS-NPs in zebrafish embryos with upregulation of genes involved in DA production (*th1*) and DA receptors *drd2a* and *drd2b* in the CNS.[118] In another study, 1 mg/L NPs inhibited AChE activity by over 40% in zebrafish larvae after 5 days of exposure.[127] In recent years, the inhibition of AChE activity induced by MPs/NPs has been also found in mammal tests. In mice, a 28-day subchronic exposure to MPs (5–20  $\mu$ m) induced inflammation and inhibited AChE activity.[72] The relative protein expression levels of AChE and microtubule-associated protein 2 were reduced and the activity of AChE was substantially inhibited, following nasal inhalation of pristine PS, PS-NH<sub>2</sub>, and PS-COOH into the brain.[10] All of the results cited suggest that MPs/NPs are capable of inhibiting AChE activity and resulting in neurotoxicity, which signifies detrimental effects on the CNS.

### 5.3. Oxidative stress

Oxidative stress arises due to a disparity between the generation of reactive oxygen species (ROS) and the effectiveness of the cellular antioxidant system.[145] As a key event in neurotoxicity, oxidative stress has been shown to be associated with the cell death in neurological diseases.[145–147] The CNS is more susceptible to oxidative harm as a result of its elevated oxygen demand and limited antioxidant capacity.[146,147] MPs/NPs may increase the levels of ROS, which stimulate lipid peroxidation and cause oxidative stress with antioxidant imbalance.[148,149] Environmental MPs/NPs were detected in 49% of frozen samples of wild fish from the Northeast Atlantic, which had significantly elevated levels of lipid peroxidation in the brain, gills, and dorsal muscle.[150] After exposure to PS-MPs, mice had increased levels of ROS and malondialdehyde (MDA) and decreased levels of glutathione

**Table 1**

Overview of the literature investigating neurotoxicity of MPs/NPs.

| Organisms                                  | Micro/nanoplastics |                         |                  | Exposure test                          |                                                                |                                 | Neurotoxic effects and possible mechanisms                                                                                                                                                                                                             | Ref.  |
|--------------------------------------------|--------------------|-------------------------|------------------|----------------------------------------|----------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Species                                    | Type               | Size                    | Characteristic   | Method of administration               | Concentration                                                  | Exposure duration               |                                                                                                                                                                                                                                                        |       |
| <b>Bivalves</b>                            |                    |                         |                  |                                        |                                                                |                                 |                                                                                                                                                                                                                                                        |       |
| Blood clam ( <i>Donax trunculus</i> )      | PS                 | 490 ± 11 nm             | Plastic sphere   | Direct exposure in aqueous suspensions | 5 mg/L                                                         | 14 days                         | By immune response and phagocytosis suppression to induce neurotransmitter disorders (GABA, DA, ACh)                                                                                                                                                   | [114] |
| Wedge clam ( <i>Scrobicularia plana</i> )  | PE / PP            | 100 - 400 µm            | Irregular shapes | Mixing plastic particles with the sand | 0.06 g/kg of sand                                              | 3 h, 1, 2, 3, 4, 7, 10, 15 days | Induced AChE inhibition and oxidative stress                                                                                                                                                                                                           | [125] |
| <b>Fish</b>                                |                    |                         |                  |                                        |                                                                |                                 |                                                                                                                                                                                                                                                        |       |
| Zebrafish embryos ( <i>Danio rerio</i> )   | PS                 | 100 nm                  | Plastic sphere   | Direct exposure in aqueous suspensions | 1, 10 and 100 µg/L                                             | From 2 to 12 hpf                | Upregulated genes related to antioxidant enzymes, changed the genes related to nervous system development and induced retinal development, stem cell differentiation, and inhibited neuronal development and locomotor activity                        | [126] |
| Zebrafish embryos ( <i>Danio rerio</i> )   | PS                 | 100 nm, 500 nm, 1000 nm | Plastic sphere   | Direct exposure in aqueous suspensions | Environmentally relevant concentration (8.6 mg/L)              | From 2 to 120 hpf               | Induced neuronal loss, motor neuron abnormalities, behavioral abnormalities, neurotransmitter disorders (GABA, 5-HT, ACh)                                                                                                                              | [100] |
| Zebrafish, embryos ( <i>Danio rerio</i> )  | PS                 | 30 nm                   | Plastic sphere   | Direct exposure in aqueous suspensions | 0.1, 0.5, and 3 mg/L                                           | From 2-120 hpf                  | Increased activity and anxiety at low doses and lethargy at high doses, induced AChE inhibition and changes in neuroendocrine gene expression                                                                                                          | [118] |
| Zebrafish, embryos ( <i>Danio rerio</i> )  | PS                 | 1, 6, 10, 25 µm         | Plastic sphere   | Direct exposure in aqueous suspensions | 0.028, 6, 27.5, 430 mg/L for the 1-, 6-, 10-, and 25-µm PS-MPs | From 6-120 hpf                  | Induced hypoactivity and seizurogenic effects, dysregulation in the cholinergic, dopaminergic, and GABAergic neurotransmission system                                                                                                                  | [109] |
| Zebrafish, larvae ( <i>Danio rerio</i> )   | PS                 | 45 µm, 50 nm            | Plastic sphere   | Direct exposure in aqueous suspensions | 1 mg/L                                                         | 5 days                          | Enhanced hypoactivity, induced oxidative stress and body length reduction, AChE inhibition                                                                                                                                                             | [127] |
| Zebrafish, larvae ( <i>Danio rerio</i> )   | PS                 | 50 nm, 100 nm           | Plastic sphere   | Direct exposure in aqueous suspensions | 0, 500, 750, 1000, 1250, 1500 mg/L                             | 96 h                            | Undermined the integrity of BBB, induced abnormal behavioral patterns, changes in electroencephalogram power spectral densities, and disturbed dopaminergic metabolites in the 50 nm PS-NP group                                                       | [123] |
| Zebrafish, juvenile ( <i>Danio rerio</i> ) | PS                 | 44 nm                   | Plastic sphere   | Direct exposure in aqueous suspensions | 1, 10 and 100 µg/L                                             | 30 days                         | Induced dysregulation of the brain-intestine-microbiota axis, altered 42 metabolites involved in neurotransmission, increased AChE activity                                                                                                            | [110] |
| Zebrafish, adult ( <i>Danio rerio</i> )    | PS                 | 70 nm                   | Plastic sphere   | Direct exposure in aqueous suspensions | 0.5, 1.5 mg/L                                                  | 7 weeks                         | Pronounced behavior alterations in locomotion activity, aggressiveness, shoal formation, and predator avoidance behavior, changed in biomarkers including AChE, DA, melatonin, GABA, 5-HT, vasopressin, kisspeptin, oxytocin, induced oxidative stress | [97]  |

(continued on next page)



Table 1 (continued)

| Organisms                                          | Micro/nanoplastics |                                 |                                     | Exposure test                          |                                            |                                         | Neurotoxic effects and possible mechanisms                                                                                                                                                 | Ref.  |
|----------------------------------------------------|--------------------|---------------------------------|-------------------------------------|----------------------------------------|--------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Species                                            | Type               | Size                            | Characteristic                      | Method of administration               | Concentration                              | Exposure duration                       |                                                                                                                                                                                            |       |
| Zebrafish, adult (Danio rerio)                     | PE                 | 40–47 $\mu\text{m}$             | Plastic sphere                      | Waterborne and foodborne exposure      | 0.1, 1 and 10 mg/L                         | For 7 days exposure and 48 h depuration | Hyperactivity after both exposure pathways, cholinergic system alteration after waterborne exposure, obvious DA and 5-HT neurotransmitter concentration changes after foodborne exposure   | [87]  |
| Goldfish (Carassius auratus)                       | PS                 | 50 nm, 500 nm, 30 $\mu\text{m}$ | Plastic sphere                      | Direct exposure in aqueous suspensions | 0.26 mg/L, 0.69 mg/L                       | 28 days                                 | Impaired olfactory-mediated behavioral responses, hampered odorant identification, action potential generation, olfactory neural signal transduction, and olfactory information processing | [89]  |
| <b>Nematodes</b><br><i>C. elegans</i>              | PS                 | 50 nm                           | Plastic sphere                      | Exposure in medium                     | 1, 10, 100, 1000 $\mu\text{g/L}$           | 72 h                                    | Induced oxidative stress, locomotion disorders, apoptosis, decrease in DA content                                                                                                          | [128] |
| <i>C. elegans</i>                                  | PS                 | 1 $\mu\text{m}$                 | Carboxyl-modified, plastic sphere   | Exposure in medium                     | 0.1, 1, 10, 100 $\mu\text{g/L}$            | 24 h                                    | Altered the expression of genes involved in neurotransmission, induced neuronal damages, and changes in the levels of glutamate, 5-HT, DA, and GABA                                        | [129] |
| <b>Mammals</b><br>C57BL/6 J mice                   | PS                 | 50 nm                           | Plastic sphere                      | Oral gavage                            | 0.5, 2.5, 10, 50 mg/kg. bw                 | 7 days                                  | Induced microglia activation and neuronal damage, undermined the integrity of BBB, induced oxidative stress, inflammatory response                                                         | [58]  |
| KM mice                                            | PS                 | 5 $\mu\text{m}$                 | Plastic sphere                      | Oral gavage                            | 0.01, 0.1, 1 mg/day                        | 4 weeks                                 | Affected the learning and memory functions, decreased the levels of ACh, induced oxidative stress                                                                                          | [81]  |
| C57BL/6 J mice                                     | PS                 | 2.5 $\mu\text{m}$               | Plastic sphere                      | Caudal vein injection                  | 1, 10, 100 mg/kg                           | Once every three days for 28 days       | Aggravated cognitive dysfunction, promoted microglia activation and pyroptosis, exacerbated CNS inflammation                                                                               | [130] |
| Mice                                               | PS                 | 50 nm                           | Plastic sphere                      | Oral gavage                            | 0.25, 2.5, 25, and 250 mg/kg bw            | 28 days                                 | Undermined the integrity of BBB, induced mitochondrial dysfunction and energy metabolism disorder, induced dopaminergic neuronal damage and PD-like neurodegeneration                      | [88]  |
| C57BL/6 J mice                                     | PS                 | 2 $\mu\text{m}$                 | Plastic sphere                      | Oral gavage                            | 0.008 mg/g or 0.016 mg/g                   | 4 or 8 weeks                            | Decreased synaptic protein levels, induced neuroinflammation, and learning and memory deficits                                                                                             | [93]  |
| BALB/c mice                                        | PS                 | 80, 100 and 200 nm              | Carboxyl-modified or amino-modified | Aerosol inhalation                     | 1 mg/day                                   | 7 days                                  | Reduced activity, induced anxiety behaviors, and AChE inhibition                                                                                                                           | [10]  |
| BALB/c mice                                        | PS                 | 0.5, 4, 10 $\mu\text{m}$        | Plastic sphere                      | Drinking water exposure                | 100 $\mu\text{g/L}$ , 1000 $\mu\text{g/L}$ | 180 days                                | Disrupted the BBB, inflammatory response, induced cognitive and memory deficits, caused lower level of dendritic spine density                                                             | [9]   |
| <b>Cell</b><br>Human neuroblastoma cells (SH-SY5Y) | PS                 | 50 nm                           | Plastic sphere                      | In culture medium                      | 2, 10, 50 $\mu\text{L/mL}$                 | 24 h                                    | Induced cytotoxicity and promoted differentiation into neuronal phenotype, induced shrinkage of neurite outgrowth,                                                                         | [131] |

(continued on next page)

Table 1 (continued)

| Organisms                                | Micro/nanoplastics |                            |                | Exposure test            |                                |                   | Neurotoxic effects and possible mechanisms                                                                                                                                                                                                                                                                                          | Ref.  |
|------------------------------------------|--------------------|----------------------------|----------------|--------------------------|--------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Species                                  | Type               | Size                       | Characteristic | Method of administration | Concentration                  | Exposure duration |                                                                                                                                                                                                                                                                                                                                     |       |
| Human microglial cells (HMC-3)           | PS                 | 0.2, 2, 10 $\mu\text{m}$   | Plastic sphere | In culture medium        | 1, 5, 10 $\mu\text{g/mL}$      | 24 h, 48 h        | morphology alteration and swelling of the nuclei, and spilling of intracellular components<br>Changed cellular morphology, induced immune responses and microglial apoptosis, altered the expression of clusters of immune response genes, immunoglobulins, and several related microRNAs                                           | [132] |
| Mouse microglia cells (BV <sub>2</sub> ) | PS                 | 2.5 $\mu\text{m}$          | Plastic sphere | In culture medium        | 1, 10, 100 mg/L                | 24 h              | Dose-dependently induced pyroptosis in BV <sub>2</sub> and RAW264.7 cells, promoted cellular expression and release of IL-1 $\beta$ , IL-6, and TNF- $\alpha$ and the expression of pyroptosis-related proteins. Excess ROS and cell cycle S phase arresting, especially at the greater concentration of smaller-sized PS particles | [130] |
| Mouse hippocampal neuronal cells (HT22)  | PS                 | 100 nm and 1 $\mu\text{m}$ | Plastic sphere | In culture medium        | 5, 25, and 75 $\mu\text{g/mL}$ | 24 h              |                                                                                                                                                                                                                                                                                                                                     | [133] |

(GSH) in brain tissue.[81] Another mouse experiment also showed that exposure to PP-MPs and/or DEHP caused an increase in ROS and MDA levels and a decrease in GSH levels in mice, suggesting that oxidative damage in the brain was induced and the joint exposure groups had additive effects.[151] Excessive levels of ROS can affect neurodevelopment and neurobehavior by negatively impacting key neuronal functions in the developing brain.[145] The increased ROS production, lipofuscin accumulation, and apoptosis indicated that oxidative stress is involved in the neurodevelopmental toxicity induced by high concentrations of PS-NPs in *C. elegans*. [128].

Antioxidant enzymes including superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidases (GPX), are crucial in safeguarding tissues against ROS injury.[152] The role of SOD is to facilitate the dismutation process of  $\text{O}_2^{\bullet-}$ , resulting in the production of  $\text{O}_2$  and  $\text{H}_2\text{O}_2$  as final products. CAT and GPX catalyze the decomposition of  $\text{H}_2\text{O}_2$  into  $\text{H}_2\text{O}$  and  $\text{O}_2$ . The activity of SOD- and CAT-based antioxidant mechanisms, which protect cells from the harmful effects of ROS, has been used frequently as biomarkers of xenobiotic-induced oxidative stress. As mentioned earlier, exposure to MPs/NPs may elevate the concentration of ROS in organisms. In response to this signal, the SOD/CAT antioxidation mechanisms would be activated. Changes in the activities and levels of CAT, SOD, and MDA were observed in the exposed group of PS-NPs, which indicated increased oxidative stress and oxidative damage in the brain of the *O. javanicus*. [142] PS-NP and/or nonylphenol exposure induced oxidative stress in the zebrafish brain by decreasing CAT activity and GSH levels, with joint exposure having more severe oxidative damage effects.[153] In addition, in terrestrial ecosystems, long-term exposure to low-density PE-MPs with high concentrations caused changes in MDA content and CAT activity, resulted in surface damage, and induced oxidative stress in earthworms (*Eisenia fetida*). [154] Meanwhile, PS-NP beads significantly upregulate antioxidant enzyme gene expression in zebrafish embryos.[126] Co-exposure of MPs with methylmercury significantly exacerbated the up-regulation of oxidative stress-related genes in zebrafish larvae.[155] Increased aspartate aminotransferase in plasma may indicate oxidative stress after joint exposure to cadmium chloride and MPs in common carp (*Cyprinus carpio*), which may stem from mitochondrial and cell membrane damage.[156].

#### 5.4. Inflammatory reactions

As the two main regulators of homeostasis, communication between the nervous system and the immune system ensures the normal functioning of organisms, of which the immune system is essential for the maintenance of normal nervous system functioning.[157] MPs/NPs are involved in disrupting immune function by inducing molecular, cellular and tissue tertiary inflammatory responses.[149] Neuroinflammation may cause irreversible damage to the CNS or lead to subsequent chronic lesions.[158].

Inflammatory responses induced by MP/NPs primarily involve the activation of pro-inflammatory cytokines, which are signaling molecules that are predominantly secreted by immune cells.[159] The existing literature unequivocally demonstrates that MPs/NPs have the capacity to modulate the expression of inflammatory response genes or the release of cytokines in *in vitro* cell line models. Genes related to immune responses and immunoglobulin were altered by 0.2 and 2  $\mu\text{m}$  PS-MP treatment in human microglial HMC-3 cells, accompanied by cell apoptosis.[132] Shan et al. documented an upregulation of proinflammatory cytokines, including tumor necrosis factor  $\alpha$  (TNF- $\alpha$ ) and interleukin-1 $\beta$  (IL-1 $\beta$ ) secretion from mouse microglia cells BV<sub>2</sub> following a 48-hour exposure to PS-NPs.[58] In the same vein, *tnfa* and *il1b* were also increased in response to 50 and 100 mg/L PS-NP exposure in astrocytes.[160] Significantly, the authors considered that the increased pro-inflammatory cytokines related to the activation of NLR family domain of pyrin containing 3 (NLRP3), playing a role in both neuronal demise and the advancement of injury.[161] The above hypothesis has also been verified in chicks' test. Yin et al. found that exposure to PS-MPs promoted the release of interleukin-18 (IL18) and IL-1 $\beta$  via activating the NLRP3-induced pyroptosis signaling pathway. [11].

Cellular reactions and tissue injuries would constitute the second and third tiers of inflammatory responses elicited by MP/NPs, respectively. Microglia, which can constitute the innate immune system and mediate neuroinflammatory processes, are essential immune system cells found in the CNS.[162] PS-NPs have been demonstrated to stimulate an inflammatory response in the hippocampus and microglia, as well as to cause neuron damage and microglial activation in the murine brain. Upon activation, these microglia exhibited enhanced motility, released

inflammatory cytokines, and migrated towards the site of injury, where they ingested injured neurons or cell debris.[9] Microglia are activated and inflammation is caused by even the slightest input, leading to neurotoxicity.[163] Kwon et al. uncovered that PS-MPs in the brain exhibited co-localization with ionized calcium-binding adaptor molecule 1, which serves as a marker for microglia, in the hippocampus, cortex, and cerebellum.[132] Immunocytochemistry labeling showed that the inactive microglia transformed into a reactive state, which was characterized by increased branching and enlarged appearance in the cerebral cortex and hippocampus when exposed to PS-NPs.[58] Activated microglia then may initiate a series of neuroinflammatory reactions and facilitate the occurrence of brain injury.[164] Inevitably, inflammatory tissue damage such as cytoplasmic vacuolation, inflammatory cell infiltration, and microthrombi appearance were determined in different *in vivo* models after exposure to MPs/NPs.[11,165].

Studies have shown that environmental factors can exacerbate inflammatory responses by disrupting the intestinal microbiota.[166] The gut microbiota is an important regulator of neurological functions.[167] Daily exposure to pollutants in humans affects the intestinal microbiota and alters the bidirectional communication between the gut and the brain, potentially inducing neurotoxicity.[168,169] PS-MPs can cause intestinal inflammation by altering the intestinal microbiota after being ingested by carp.[170] Meanwhile, zebrafish intestinal inflammation caused by PS-NPs was closely associated with dysregulation within the brain-gut-microbiota axis.[110] Pronounced inflammatory alterations and tissue degradation were observed in intestinal, liver and kidney sections of *O. javanicus* after chronic exposure to PS-MPs.[142].

### 5.5. Mitochondrial dysfunction

Mitochondria possess a wide range of biological activities and play a role in regulating adenosine triphosphate (ATP) synthesis, apoptosis, lipid metabolism, calcium signaling, and other biological processes.[171] Impaired functioning of mitochondria leads to a decrease in ATP, causing the enzymes involved in the electron transport chain to stop working. This results in the production of ROS and pro-inflammatory cytokines,[99] contributing to the pathogenesis of neurological disorders.

Structural damages such as the removal of the mitochondrial matrix and cristae, swelling and rupture of mitochondria, and vacuolated mitochondria have been widely reported after exposure to MPs/NPs.[172–174] These damages may trigger off a series of dysfunctions in mitochondrial function, including abnormalities in mitochondrial dynamics, electron transport, mitophagy, and other processes.

For instance, an *in vitro* assay using differentiated SH-SY5Y cells demonstrated that PS-NPs induced notable mitochondrial harm, defined by changes in structure, decreased mitochondrial membrane potential, and reduced ATP generation. PS-NPs triggered excessive mitophagy and led to further deterioration of mitochondrial function and ultimate death of dopaminergic neurons.[175] Further research found that PS-NPs at concentrations over 100 mg/L had a notable impact on neurotoxic effects evidenced by the disruption of mitochondrial function, as revealed by elevated levels of intracellular  $\text{Ca}^{2+}$  and reduced levels of mitochondrial membrane potential and ATP. Furthermore, PS-NPs stimulated the mitochondrial apoptotic pathway and facilitated cell death. Liang et al. utilized single-nucleus RNA sequencing to analyze single-cell transcriptional alterations in mouse brains. Results showed PS-NPs may induce mitochondrial dysfunction and energy metabolism disorders predominantly in excitatory neurons.[88] Additionally, Yin et al. demonstrated that PS-MPs led to changes in mitochondrial dynamics-related metrics including proteins involved in mitochondrial fission and fusion in a dose-dependent manner, providing an abundance of evidence that mitochondrial dynamic homeostasis participates in the NP-induced neurotoxicity.[11] Another *in vivo* study based on zebrafish also found mitochondrial damage in NP-exposed brain, manifested as lower mitochondrial membrane potential, reduced mtDNA copy number

and changed mRNA expression levels of genes related to mitochondrial biosynthesis, fusion and division in the brain.[174].

There is also a correlation between particle size and mitochondrial dysfunction induced by MPs/NPs, with smaller particles producing greater effects.[176] Trevisan et al. provided evidence that supports this logic by highlighting that NPs, in comparison to MPs, have a higher tendency to concentrate in the lipid-rich environment of mitochondria.[177] This accumulation leads to a disruption in mitochondrial energy generation in zebrafish. In a similar fashion, photolytic PLA MPs were more significant in interfering with mitochondrial function in zebrafish larvae than their original PLA counterparts, due to the fact that photodegradation reduced the size of the MPs and produced NPs.[173] In addition, degraded MP fragments had a rough surface with numerous fissures and openings, allowing them to remain in the body for a prolonged period of time and exert more potent toxic effects than pristine PLA with a smooth surface.[178].

### 5.6. Impairment in autophagy-lysosome pathway

Autophagy, an essential metabolic process in mammalian cells, eliminates aberrant cellular proteins and may therefore have a neuro-protective effect.[179] Lysosomes, which are composed of vesicles encapsulating hydrolytic enzymes and are membrane-bound, are the most crucial organelles in the concluding step of autophagy.[180] Autophagosomes must undergo fusion with lysosomes during autophagy in order to degrade protein aggregates; these aggregates must be eliminated via the autophagic pathway; failure to do so results in their toxicity.[181] Recent reports found that MPs/NPs can induce neurotoxicity by disrupting the normal regulation of autophagy-lysosome pathway. Particles with specific features can either trigger the autophagy pathway and translocate to lysosomes, or inhibit autophagic flux even when autophagy is activated.[182,183] Another study aiming to determine the autophagy effect of NPs also found that 100 nm PS-NPs triggered the onset of autophagy and the creation of autophagosomes, but hindered autophagic flux in human umbilical vein endothelial cells.[184].

The potential influence of particle characteristics, including size and surface charge, on the autophagy-lysosome pathway is a matter of concern. For instance, Yang and Wang discovered that 100 nm PS-NPs caused an increase in alkalinity in cell lysosomal compartments, confirming that smaller-sized NPs activated the lysosomal proton pump.[185] Moreover, 1000 nm PS-NPs caused acidification in the lysosomal compartment but did not noticeably disrupt the structure of the lysosomes. PS-NPs with a neutral or negative charge ( $\text{COOH}^-$ ) increased the autophagic flux or the removal of cellular waste, whereas PS-NPs with a positive charge ( $\text{NH}_2^+$ ) caused problems with lysosomes and blocked the autophagic flux.[183,186] Fig. 3.

## 6. Concluding remarks and perspectives

The pollution of MPs/NPs has pervaded the environment and raised concerns over their effects on human health. Mounting evidence indicates that being exposed to MPs/NPs contributes to the incidence and progression of neurological illnesses. By comprehensively summarizing the existing research, we critically examined the present understanding of MPs/NPs on environmental risks, pathways of exposure, deleterious effects on the brain, and the underlying mechanisms. This review offers valuable insights into comprehending the susceptibility to neurological illnesses caused by MPs/NPs. Knowledge gaps and recommendations for future research on the effects of MPs/NPs on brain health are listed as follows:

- (1) Upon entering the human body, MPs/NPs could be transferred to diverse tissues, including the brain. Understanding the distributions of MPs/NPs in the human body is the utmost priority in exploring their potentially harmful consequences. Despite efforts

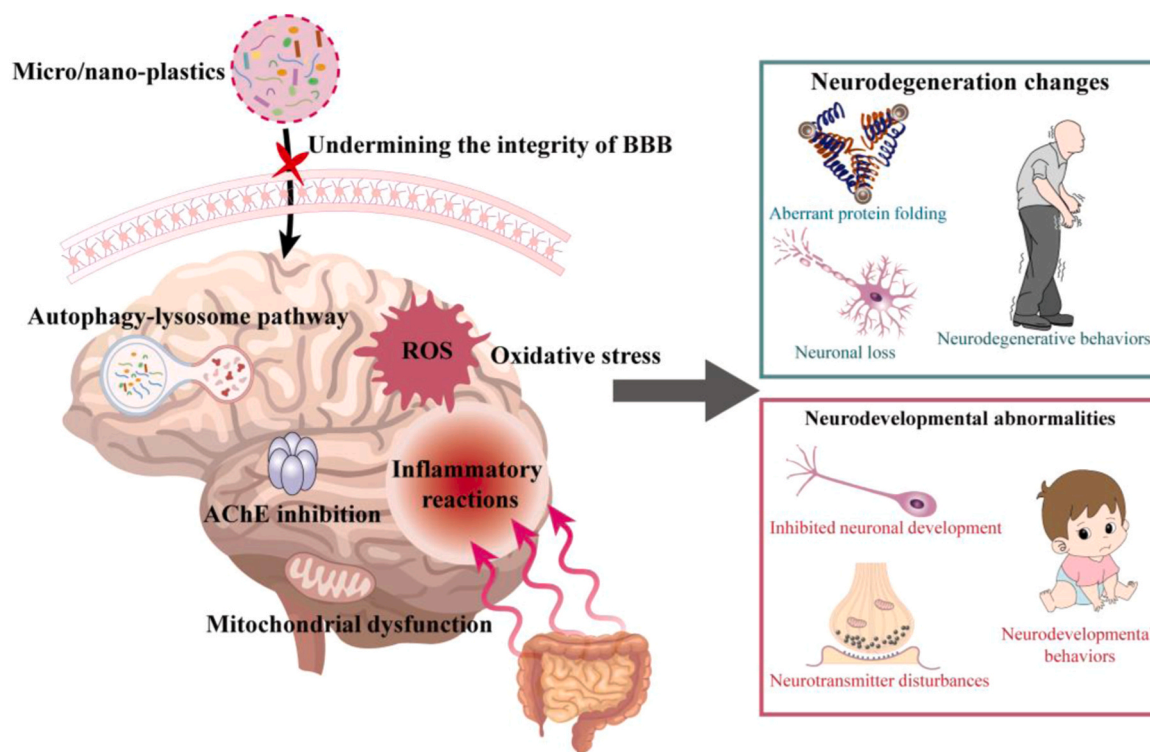


Fig. 3. Risks of neurological diseases induced by MPs/NPs and possible mechanisms.

in research to elucidate the “absorption-distribution-metabolism-excretion” process of MPs/NPs, the precise migration and transformation pathways through which they reach individual tissues remain unknown. CNS is a potential toxic target organ of MPs/NPs, whereas explanations for MPs/NPs entering the brain are mostly based on nanoparticle studies. More studies are warranted to validate the mechanisms by which internal plastics are transferred to the brain.

- (2) The environment contains several types of MPs/NPs, such as PE, PS, PP, and PET. These plastics have complex sizes and shapes including fibre, flake, sphere, fragment and irregular shapes, while existing toxicological evidence regarding the assessment of MP/NP toxicity is mostly modeled with commercial PS microplastic balls. Moreover, most neurotoxicity assays set concentrations much higher than the actual environmental levels, which is of little significance for toxicity assessment. Therefore, real evaluation data will be needed especially considering particle characteristics and exposure conditions.
- (3) Due to the attribution of environmental behaviours, MPs/NPs frequently undergo the adsorption of diverse environmental pollutants in the real environment, encompassing organic matters and metal pollutants. Once these MPs/NPs, which are combined with environmental pollutants, penetrate the brain, they will probably exert a pronounced toxicity due to synergistic effects. Nevertheless, there is a dearth of evaluation data about the neurotoxicity of plastic composites. To this end, further investigations regarding the joint effects of MPs/NPs in combination with the absorbed pollutants are essential.
- (4) Despite an increase in toxicological studies on the biotoxicity of MPs/NPs in recent years, there is a scarcity of research examining the toxicological consequences of inhaling airborne MPs/NPs on the human brain. As one of the main exposure ways of MPs/NPs in the human body, inhalation of plastics has become a hot spot recently, and further experiments will be needed to unravel the effects and mechanisms of inhalable MPs/NPs contributing to neurological disorders.

- (5) Researches have demonstrated that the neurotoxic effects induced by MPs/NPs traverse numerous regulatory pathways, including oxidative stress, inhibition of AChE, and immunoinflammatory response, among others. Nevertheless, the exact mechanism by which MPs/NPs elicit neurotoxicity remains unknown. It is also uncertain what the logical connection is between the initiating events, molecular key events and the development of brain disorders triggered by MPs/NPs. Subsequent investigations will be suggested to incorporate a mechanism-based framework, such as the adverse outcome pathway, to formally and technically classify and link these detrimental effects on neurological diseases.

### Environmental implications

Environmental MP/NP pollution poses a great threat to human health, and the emergence and development of many diseases are linked to the rising plastic contamination. This systematic review summarizes the risks of environmental MPs/NPs, pathways of MPs/NPs entering into the brain, neurotoxic effects and possible mechanisms with respect to the development of neurological diseases. The review is the first paper to summarize the neurotoxic effects of environmental MPs/NPs from the standpoint of human brain health, helping us better understand the dangers of MPs/NPs and the pathogenesis of diseases from the perspective of environmental pollution.

### CRediT authorship contribution statement

**Shuang Liu:** Methodology, Investigation, Data curation, and Writing - original draft. **Yinling He:** Methodology, Investigation, Data curation, and Writing - original draft. **Jia Yia:** Visualization, and Formal analysis. **Qingqing Zhu:** Formal analysis, and Writing - review & editing. **Chunyang Liao:** Conceptualization, Investigation, Project administration, Resources, Supervision, and Writing - review & editing. **Guibin Jiang:** Resources, and Supervision. All authors contributed extensively to writing and revising the paper.



## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2024.134054.

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